

Homo Deus: A Brief History of Tomorrow by Yuval Noah Harari

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For the first time in history, more people die today from eating too much than from eating too little; more people die from old age than from infectious diseases; and more people commit suicide than are killed by soldiers, terrorists and criminals combined. In the early twenty-first century, the average human is far more likely to die from bingeing at McDonald's than from drought, Ebola or an al-Qaeda attack.[Read more at location 251](#)

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During the last hundred years, technological, economic and political developments have created an increasingly robust safety net separating humankind from the biological poverty line. Mass famines still strike some areas from time to time, but they are exceptional, and they are almost always caused by human politics rather than by natural catastrophes. In most parts of the planet, even if a person has lost his job and all of his possessions, he is unlikely to die from hunger. Private insurance schemes, government agencies and international NGOs may not rescue him from poverty, but they will provide him with enough daily calories to survive. On the collective level, the global trade network turns droughts and floods into business opportunities, and makes it possible to overcome food shortages quickly and cheaply.[Read more at location 287](#)

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Since 1974 hundreds of millions of Chinese have been lifted out of poverty, and though hundreds of millions more still suffer greatly from privation and malnutrition, for the first time in its recorded history China is now free from famine.[Read more at location 306](#)

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Whereas in ancient agricultural societies human violence caused about 15 per cent of all deaths, during the twentieth century violence caused only 5 per cent of deaths, and in the early twenty-first century it is responsible for about 1 per cent of global mortality.[Read more at location 450](#)

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The global economy has been transformed from a material-based economy into a knowledge-based economy. Previously the main sources of wealth were material assets such as gold mines, wheat fields and oil wells. Today the main source of wealth is knowledge. And whereas you can conquer oil fields through war, you cannot acquire knowledge that way. Hence as knowledge became the most important economic resource, the profitability of war declined and wars became increasingly restricted to those parts of the world – such as the Middle East and Central Africa – where the economies are still old-fashioned material-based economies.[Read more at location 458](#)

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In essence, terrorism is a show. Terrorists stage a terrifying spectacle of violence that captures our imagination and makes us feel as if we are sliding back into medieval chaos. Consequently states often feel obliged to react to the theatre of terrorism with a show of security, orchestrating immense displays of force, such as the persecution of entire populations or the invasion of foreign countries. In most cases, this overreaction to terrorism poses a far greater threat to our security than the terrorists themselves.[Read more at location 506](#)

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Acknowledging our past achievements sends a message of hope and responsibility, encouraging us to make even greater efforts in the future. Given our twentieth-century accomplishments, if people continue to suffer from famine, plague and war, we cannot blame it on nature or on God. It is within our power to make things better and to reduce the incidence of suffering even further.[Read more at location 526](#)

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History does not tolerate a vacuum. If incidences of famine, plague and war are decreasing, something is bound to take their place on the human agenda. We had better think very carefully what it is going to be. Otherwise, we might gain complete victory in the old battlefields only to be caught completely unaware on entirely new fronts. What are the projects that will replace famine, plague and war at the top of the human agenda in the twenty-first century?[Read more at location 529](#)

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One central project will be to protect humankind and the planet as a whole from the dangers inherent in our own power.[Read more at location 533](#)

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When the moment comes to choose between economic growth and ecological stability, politicians, CEOs and voters almost always prefer growth. In the twenty-first century, we shall have to do better if we are to avoid catastrophe.[Read more at location 538](#)

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Success breeds ambition, and our recent achievements are now pushing humankind to set itself even more daring goals. Having secured unprecedented levels of prosperity, health and harmony, and given our past record and our current values, humanity's next targets are likely to be immortality, happiness and divinity.[Read more at location 545](#)

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In the twenty-first century humans are likely to make a serious bid for immortality. Struggling against old age and death will merely carry on the time-honoured fight against famine and disease, and manifest the supreme value of contemporary culture: the worth of human life. We are constantly reminded that human life is the most sacred thing in the universe.[Read more at location 551](#)

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Modern science and modern culture have an entirely different take on life and death. They don't think of death as a metaphysical mystery, and they certainly don't view death as the source of life's meaning. Rather, for modern people death is a technical problem that we can and should solve.[Read more at location 564](#)

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If traditionally death was the speciality of priests and theologians, now the engineers are taking over. We can kill the cancerous cells with chemotherapy or nano-robots. We can exterminate the germs in the lungs with antibiotics. If the heart stops pumping, we can reinvigorate it with medicines and electric shocks – and if that doesn't work, we can implant a new heart. True, at present we don't have solutions to all technical problems. But this is precisely why we invest so much time and money in researching cancer, germs, genetics and nanotechnology.[Read more at location 581](#)

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The breakneck development of fields such as genetic engineering, regenerative medicine and nanotechnology fosters ever more optimistic prophecies. Some experts believe that humans will overcome death by 2200, others say 2100. Kurzweil and de Grey are even more sanguine. They maintain that anyone possessing a healthy body and a healthy bank account in 2050 will have a serious shot at immortality by cheating death a decade at a time.[Read more at location 613](#)

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Although average life expectancy has doubled over the last hundred years, it is unwarranted to extrapolate and conclude that we can double it again to 150 in the coming century. In 1900 global life expectancy was no higher than forty because many people died young from malnutrition, infectious diseases and violence. Yet those who escaped famine, plague and war could live well into their seventies and eighties, which is the natural life span of Homo sapiens.[Read more at location 650](#)

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Once people think (with or without good reason) that they have a serious chance of escaping death, the desire for life will refuse to go on pulling the rickety wagon of art, ideology and religion, and will sweep forward like an avalanche.[Read more at location 684](#)

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In ancient Greece the philosopher Epicurus explained that worshipping gods is a waste of time, that there is no existence after death, and that happiness is the sole purpose of life. Most people in ancient times rejected Epicureanism, but today it has become the default view.[Read more at location 693](#)

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People increasingly believe that the immense systems established more than a century ago to strengthen the nation should actually serve the happiness and well-being of individual citizens. We are not here to serve the state – it is here to serve us. The right to the pursuit of happiness, originally envisaged as a restraint on state power, has imperceptibly morphed into the right to happiness – as if human beings have a natural right to be happy, and anything which makes us dissatisfied is a violation of our basic human rights, so the state should do something about it.[Read more at location 729](#)

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Being happy doesn't come easy. Despite our unprecedented achievements in the last few decades, it is far from obvious that contemporary people are significantly more

satisfied than their ancestors in bygone years. Indeed, it is an ominous sign that despite higher prosperity, comfort and security, the rate of suicide in the developed world is also much higher than in traditional societies.[Read more at location 747](#)

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The glass ceiling of happiness is held in place by two stout pillars, one psychological, the other biological. On the psychological level, happiness depends on expectations rather than objective conditions. We don't become satisfied by leading a peaceful and prosperous existence. Rather, we become satisfied when reality matches our expectations. The bad news is that as conditions improve, expectations balloon. Dramatic improvements in conditions, as humankind has experienced in recent decades, translate into greater expectations rather than greater contentment. If we don't do something about this, our future achievements too might leave us as dissatisfied as ever. On the biological level, both our expectations and our happiness are determined by our biochemistry, rather than by our economic, social or political situation. According to Epicurus, we are happy when we feel pleasant sensations and are free from unpleasant ones. Jeremy Bentham similarly maintained that nature gave dominion over man to two masters – pleasure and pain – and they alone determine everything we do, say and think. Bentham's successor, John Stuart Mill, explained that happiness is nothing but pleasure and freedom from pain, and that beyond pleasure and pain there is no good and no evil.[Read more at location 778](#)

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Perhaps the key to happiness is neither the race nor the gold medal, but rather combining the right doses of excitement and tranquillity; but most of us tend to jump all the way from stress to boredom and back, remaining as discontented with one as with the other.[Read more at location 844](#)

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The state hopes to regulate the biochemical pursuit of happiness, separating 'bad' manipulations from 'good' ones. The principle is clear: biochemical manipulations that strengthen political stability, social order and economic growth are allowed and even encouraged (e.g. those that calm hyperactive kids in school, or drive anxious soldiers forward into battle). Manipulations that threaten stability and growth are banned. But each year new drugs are born in the research labs of universities, pharmaceutical companies and criminal organisations, and the needs of the state and the market also keep changing. As the biochemical pursuit of happiness accelerates, so it will reshape politics, society and economics, and it will become ever harder to bring it under control.[Read more at location 875](#)

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Some 2,300 years ago Epicurus warned his disciples that immoderate pursuit of pleasure is likely to make them miserable rather than happy. A couple of centuries earlier Buddha had made an even more radical claim, teaching that the pursuit of pleasant sensations is in fact the very root of suffering. Such sensations are just ephemeral and meaningless vibrations. Even when we experience them, we don't react to them with contentment; rather, we just crave for more. Hence no matter how many blissful or exciting sensations I may experience, they will never satisfy me. If I identify happiness with fleeting pleasant sensations, and crave to experience more and more of them, I have no choice but to pursue them constantly. When I finally get them, they quickly disappear, and because the mere memory of past pleasures will not satisfy me, I have to start all over again. Even if I continue this pursuit for decades, it will never bring me any lasting achievement; on the contrary, the more I crave these pleasant sensations, the more stressed and dissatisfied I will become. To attain real happiness, humans need to slow down the pursuit of pleasant sensations, not accelerate it.[Read more at location 889](#)

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In seeking bliss and immortality humans are in fact trying to upgrade themselves into gods. Not just because these are divine qualities, but because in order to overcome old age and misery humans will first have to acquire godlike control of their own biological substratum. If we ever have the power to engineer death and pain out of our system, that same power will probably be sufficient to engineer our system in almost any manner we like, and manipulate our organs, emotions and intelligence in myriad ways. You could buy for yourself the strength of Hercules, the sensuality of Aphrodite, the wisdom of Athena or the madness of Dionysus if that is what you are into.[Read more at location 914](#)

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The upgrading of humans into gods may follow any of three paths: biological engineering, cyborg engineering and the engineering of non-organic beings.[Read more at](#)

For thousands of years history was full of technological, economic, social and political upheavals. Yet one thing remained constant: humanity itself. Our tools and institutions are very different from those of biblical times, but the deep structures of the human mind remain the same. This is why we can still find ourselves between the pages of the Bible, in the writings of Confucius or within the tragedies of Sophocles and Euripides. These classics were created by humans just like us, hence we feel that they talk about us. In modern theatre productions, Oedipus, Hamlet and Othello may wear jeans and T-shirts and have Facebook accounts, but their emotional conflicts are the same as in the original play. However, once technology enables us to re-engineer human minds, Homo sapiens will disappear, human history will come to an end and a completely new kind of process will begin, which people like you and me cannot comprehend.[Read more at location 959](#)

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Though the details are therefore obscure, we can nevertheless be sure about the general direction of history. In the twenty-first century, the third big project of humankind will be to acquire for us divine powers of creation and destruction, and upgrade Homo sapiens into Homo deus. This third project obviously subsumes the first two projects, and is fuelled by them. We want the ability to re-engineer our bodies and minds in order, above all, to escape old age, death and misery, but once we have it, who knows what else we might do with such ability? So we may well think of the new human agenda as consisting really of only one project (with many branches): attaining divinity.[Read more at location 970](#)

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An economy built on everlasting growth needs endless projects – just like the quests for immortality, bliss and divinity. Well, if we need limitless projects, why not settle for bliss and immortality, and at least put aside the frightening quest for superhuman powers? Because it is inextricable from the other two. When you develop bionic legs that enable paraplegics to walk again, you can also use the same technology to upgrade healthy people. When you discover how to stop memory loss among older people, the same treatments might enhance the memory of the young.[Read more at location 1052](#)

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Once you achieve a momentous breakthrough, you cannot restrict its use to healing and completely forbid using it for upgrading.[Read more at location 1109](#)

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Precisely because we have some choice regarding the use of new technologies, we had better understand what is happening and make up our minds about it before it makes up our minds for us.[Read more at location 1115](#)

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This is the paradox of historical knowledge. Knowledge that does not change behaviour is useless. But knowledge that changes behaviour quickly loses its relevance. The more data we have and the better we understand history, the faster history alters its course, and the faster our knowledge becomes outdated.[Read more at location 1159](#)

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Historians study the past not in order to repeat it, but in order to be liberated from it. Each and every one of us has been born into a given historical reality, ruled by particular norms and values, and managed by a unique economic and political system. We take this reality for granted, thinking it is natural, inevitable and immutable. We forget that our world was created by an accidental chain of events, and that history shaped not only our technology, politics and society, but also our thoughts, fears and dreams. The cold hand of the past emerges from the grave of our ancestors, grips us by the neck and directs our gaze towards a single future. We have felt that grip from the moment we were born, so we assume that it is a natural and inescapable part of who we are. Therefore we seldom try to shake ourselves free, and envision alternative futures. Studying history aims to loosen the grip of the past. It enables us to turn our head this way and that, and begin to notice possibilities that our ancestors could not imagine, or didn't want us to imagine. By observing the accidental chain of events that led us here, we realise how our very thoughts and dreams took shape – and we can begin to think and dream differently. Studying history will not tell us what to choose, but at least it gives us more options.[Read more at location 1181](#)

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Despite the technological hubris of these dreams, ideologically they are old news. For 300 years the world has been dominated by humanism, which sanctifies the life, happiness and power of Homo sapiens. The attempt to gain immortality, bliss and divinity merely takes the long-standing humanist ideals to their logical conclusion.[Read more at location 1260](#)

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Yet the rise of humanism also contains the seeds of its downfall. While the attempt to upgrade humans into gods takes humanism to its logical conclusion, it simultaneously exposes humanism's inherent flaws. If you start with a flawed ideal, you often appreciate its defects only when the ideal is close to realisation. We can already see this process at work in geriatric hospital wards. Due to an uncompromising humanist belief in the sanctity of human life, we keep people alive till they reach such a pitiful state that we are forced to ask, 'What exactly is so sacred here?' Due to similar humanist beliefs, in the twenty-first century we are likely to push humankind as a whole beyond its limits. The same technologies that can upgrade humans into gods might also make humans irrelevant. For example, computers powerful enough to understand and overcome the mechanisms of ageing and death will probably also be powerful enough to replace humans in any and all tasks.[Read more at location 1265](#)

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At present, more than 90 per cent of the large animals of the world (i.e. those weighing more than a few kilograms) are either humans or domesticated animals.[Read more at location 1332](#)

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This is the basic lesson of evolutionary psychology: a need shaped thousands of generations ago continues to be felt subjectively even if it is no longer necessary for survival and reproduction in the present. Tragically, the Agricultural Revolution gave humans the power to ensure the survival and reproduction of domesticated animals while ignoring their subjective needs.[Read more at location 1495](#)

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Whereas hunter-gatherers were seldom aware of the damage they inflicted on the ecosystem, farmers knew perfectly well what they were doing. They knew they were exploiting domesticated animals and subjugating them to human desires and whims. They justified their actions in the name of new theist religions, which mushroomed and spread in the wake of the Agricultural Revolution. Theist religions maintained that the universe is ruled by a group of great gods – or perhaps by a single capital 'G' God. We don't normally associate this idea with agriculture, but at least in their beginnings theist religions were an agricultural enterprise. The theology, mythology and liturgy of religions such as Judaism, Hinduism and Christianity revolved at first around the relationship between humans, domesticated plants and farm animals.[Read more at location 1623](#)

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The rise of modern science and industry brought about the next revolution in human-animal relations. During the Agricultural Revolution humankind silenced animals and plants, and turned the animist grand opera into a dialogue between man and gods. During the Scientific Revolution humankind silenced the gods too. The world was now a one-man show. Humankind stood alone on an empty stage, talking to itself, negotiating with no one and acquiring enormous powers without any obligations. Having deciphered the mute laws of physics, chemistry and biology, humankind now does with them as it pleases.[Read more at location 1726](#)

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Whereas the Agricultural Revolution gave rise to theist religions, the Scientific Revolution gave birth to humanist religions, in which humans replaced gods. While theists worship theos (Greek for 'god'), humanists worship humans. The founding idea of humanist religions such as liberalism, communism and Nazism is that Homo sapiens has some unique and sacred essence that is the source of all meaning and authority in the universe. Everything that happens in the cosmos is judged to be good or bad according to its impact on Homo sapiens.[Read more at location 1752](#)

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With the help of vaccinations, medications, hormones, pesticides, central air-conditioning systems and automatic feeders, it is now possible to pack tens of thousands of pigs, cows or chickens into neat rows of cramped cages, and produce meat, milk and eggs with unprecedented efficiency. In recent years, as people began to rethink human-animal relations, such practices have come under increasing criticism. We are suddenly showing unprecedented interest in the fate of so-called lower life

forms, perhaps because we are about to become one. If and when computer programs attain superhuman intelligence and unprecedented power, should we begin valuing these programs more than we value humans? Would it be okay, for example, for an artificial intelligence to exploit humans and even kill them to further its own needs and desires? If it should never be allowed to do that, despite its superior intelligence and power, why is it ethical for humans to exploit and kill pigs? Do humans have some magical spark, in addition to higher intelligence and greater power, which distinguishes them from pigs, chickens, chimpanzees and computer programs alike? If yes, where did that spark come from, and why are we certain that an AI could never acquire it? If there is no such spark, would there be any reason to continue assigning special value to human life even after computers surpass humans in intelligence and power? Indeed, what exactly is it about humans that make us so intelligent and powerful in the first place, and how likely is it that non-human entities will ever rival and surpass us?[Read more at location 1767](#)

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When we privilege human children over piglets, we want to believe that this reflects something deeper than the ecological balance of power. We want to believe that human lives really are superior in some fundamental way. We Sapiens love telling ourselves that we enjoy some magical quality that not only accounts for our immense power, but also gives moral justification for our privileged status. What is this unique human spark? The traditional monotheist answer is that only Sapiens have eternal souls. Whereas the body decays and rots, the soul journeys on towards salvation or damnation, and will experience either everlasting joy in paradise or an eternity of misery in hell. Since pigs and other animals have no soul, they don't take part in this cosmic drama. They live only for a few years, and then die and fade into nothingness. We should therefore care far more about eternal human souls than about ephemeral pigs.[Read more at location 1791](#)

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The belief that humans have eternal souls whereas animals are just evanescent bodies is a central pillar of our legal, political and economic system. It explains why, for example, it is perfectly okay for humans to kill animals for food, or even just for the fun of it. However, our latest scientific discoveries flatly contradict this monotheist myth. True, laboratory experiments confirm the accuracy of one part of the myth: just as monotheist religions say, animals have no souls. All the careful studies and painstaking examinations have failed to discover any trace of a soul in pigs, rats or rhesus monkeys. Alas, the same laboratory experiments undermine the second and far more important part of the monotheist myth, namely, that humans do have a soul. Scientists have subjected Homo sapiens to tens of thousands of bizarre experiments, and looked into every nook in our hearts and every cranny in our brains. But they have so far discovered no magical spark. There is zero scientific evidence that in contrast to pigs, Sapiens have souls.[Read more at location 1800](#)

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Darwin has deprived us of our souls. If you really understand the theory of evolution, you understand that there is no soul. This is a terrifying thought not only to devout Christians and Muslims, but also to many secular people who don't hold any clear religious dogma, but nevertheless want to believe that each human possesses an eternal individual essence that remains unchanged throughout life, and can survive even death intact.[Read more at location 1831](#)

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The theory of evolution rejects the idea that my true self is some indivisible, immutable and potentially eternal essence. According to the theory of evolution, all biological entities – from elephants and oak trees to cells and DNA molecules – are composed of smaller and simpler parts that ceaselessly combine and separate. Elephants and cells have evolved gradually, as a result of new combinations and splits. Something that cannot be divided or changed cannot have come into existence through natural selection.[Read more at location 1839](#)

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Another story employed to justify human superiority says that of all the animals on earth, only Homo sapiens has a conscious mind. Mind is something very different from soul. The mind isn't some mystical eternal entity. Nor is it an organ such as the eye or the brain. Rather, the mind is a flow of subjective experiences, such as pain, pleasure, anger and love. These mental experiences are made of interlinked sensations, emotions and thoughts, which flash for a brief moment, and immediately disappear.[Read more at location 1869](#)

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What exactly are the conscious experiences that constitute the flow of the mind? Every subjective experience has two fundamental characteristics: sensation and desire. Robots and computers have no consciousness because despite their myriad abilities they feel nothing and crave nothing.[Read more at location 1879](#)

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In the seventeenth century Descartes maintained that only humans feel and crave, whereas all other animals are mindless automata, akin to a robot or a vending machine.[Read more at location 1893](#)

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The most common explanation points out that the brain is a highly complex system, with more than 80 billion neurons connected into numerous intricate webs. When billions of neurons send billions of electric signals back and forth, subjective experiences emerge. Even though the sending and receiving of each electric signal is a simple biochemical phenomenon, the interaction among all these signals creates something far more complex – the stream of consciousness. We observe the same dynamic in many other fields. The movement of a single car is a simple action, but when millions of cars move and interact simultaneously, traffic jams emerge.[Read more at location 1921](#)

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Yet this explanation explains nothing. It merely affirms that the problem is very complicated. It does not offer any insight into how one kind of phenomenon (billions of electric signals moving from here to there) creates a very different kind of phenomenon (subjective experiences of anger or love). The analogy to other complex processes such as traffic jams and economic crises is flawed.[Read more at location 1927](#)

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Philosophers have encapsulated this riddle in a trick question: what happens in the mind that doesn't happen in the brain? If nothing happens in the mind except what happens in our massive network of neurons – then why do we need the mind? If something does indeed happen in the mind over and above what happens in the neural network – where the hell does it happen?[Read more at location 1980](#)

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Despite the vast knowledge we have gained in the fields of mathematics and computer science, none of the data-processing systems we have created needs subjective experiences in order to function, and none feels pain, pleasure, anger or love.[Read more at location 1998](#)

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Some scientists – such as Daniel Dennett and Stanislas Dehaene – argue that all relevant questions can be answered by studying brain activities, without any recourse to subjective experiences. So scientists can safely delete 'mind', 'consciousness' and 'subjective experiences' from their vocabulary and articles. However, as we shall see in the following chapters, the whole edifice of modern politics and ethics is built upon subjective experiences, and few ethical dilemmas can be solved by referring strictly to brain activities.[Read more at location 2037](#)

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Some scientists concede that consciousness is real and may actually have great moral and political value, but that it fulfils no biological function whatsoever. Consciousness is the biologically useless by-product of certain brain processes. Jet engines roar loudly, but the noise doesn't propel the aeroplane forward.[Read more at location 2045](#)

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The Turing Test was invented in 1950 by the British mathematician Alan Turing, one of the fathers of the computer age. Turing was also a gay man in a period when homosexuality was illegal in Britain. In 1952 he was convicted of committing homosexual acts and forced to undergo chemical castration. Two years later he committed suicide. The Turing Test is simply a replication of a mundane test every gay man had to undergo in 1950 Britain: can you pass for a straight man? Turing knew from personal experience that it didn't matter who you really were – it mattered only what others thought about you. According to Turing, in the future computers would be just like gay men in the 1950s. It won't matter whether computers will actually be conscious or not. It will matter only what people think about it.[Read more at location 2105](#)

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On 7 July 2012 leading experts in neurobiology and the cognitive sciences gathered at

the University of Cambridge, and signed the Cambridge Declaration on Consciousness, which says that 'Convergent evidence indicates that non-human animals have the neuroanatomical, neurochemical and neurophysiological substrates of conscious states along with the capacity to exhibit intentional behaviours. Consequently, the weight of evidence indicates that humans are not unique in possessing the neurological substrates that generate consciousness. Non-human animals, including all mammals and birds, and many other creatures, including octopuses, also possess these neurological substrates.'⁹ This declaration stops short of saying that other animals are conscious, because we still lack the smoking gun. But it does shift the burden of proof to those who think otherwise.[Read more at location 2132](#)

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In essence, we humans are not that different from rats, dogs, dolphins or chimpanzees. Like them, we too have no soul. Like us, they too have consciousness and a complex world of sensations and emotions. Of course, every animal has its unique traits and talents. Humans too have their special gifts. We shouldn't humanise animals needlessly, imagining that they are just a furrer version of ourselves. This is not only bad science, but it also prevents us from understanding and valuing other animals on their terms.[Read more at location 2236](#)

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The crucial factor in our conquest of the world was our ability to connect many humans to one another.¹⁸ Humans nowadays completely dominate the planet not because the individual human is far smarter and more nimble-fingered than the individual chimp or wolf, but because *Homo sapiens* is the only species on earth capable of co-operating flexibly in large numbers. Intelligence and toolmaking were obviously very important as well. But if humans had not learned to cooperate flexibly in large numbers, our crafty brains and deft hands would still be splitting flint stones rather than uranium atoms. If cooperation is the key, how come the ants and bees did not beat us to the nuclear bomb even though they learned to cooperate en masse millions of years before us? Because their cooperation lacks flexibility. Bees cooperate in very sophisticated ways, but they cannot reinvent their social system overnight. If a hive faces a new threat or a new opportunity, the bees cannot, for example, guillotine the queen and establish a republic. Social mammals such as elephants and chimpanzees cooperate far more flexibly than bees, but they do so only with small numbers of friends and family members. Their cooperation is based on personal acquaintance. If I am a chimpanzee and you are a chimpanzee and I want to cooperate with you, I must know you personally: what kind of chimp are you? Are you a nice chimp? Are you an evil chimp? How can I cooperate with you if I don't know you? To the best of our knowledge, only *Sapiens* can cooperate in very flexible ways with countless numbers of strangers. This concrete capability – rather than an eternal soul or some unique kind of consciousness – explains our mastery of planet Earth.[Read more at location 2284](#)

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A similar fate befell the Egyptian Revolution of 2011. What television did in 1989, Facebook and Twitter did in 2011. The new media helped the masses coordinate their activities, so that thousands of people flooded the streets and squares at the right moment and toppled the Mubarak regime. However, it is one thing to bring 100,000 people to Tahrir Square, and quite another to get a grip on the political machinery, shake the right hands in the right back rooms and run a country effectively. Consequently, when Mubarak stepped down the demonstrators could not fill the vacuum. Egypt had only two institutions sufficiently organised to rule the country: the army and the Muslim Brotherhood. Hence the revolution was hijacked first by the Brotherhood, and eventually by the army.[Read more at location 2373](#)

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Research indicates that *Sapiens* just can't have intimate relations (whether hostile or amorous) with more than 150 individuals.²² Whatever enables humans to organise mass-cooperation networks, it isn't intimate relations. This is bad news for psychologists, sociologists, economists and others who try to decipher human society through laboratory experiments. For both organisational and financial reasons, the vast majority of experiments are conducted either on individuals or on small groups of participants. Yet it is risky to extrapolate from small-group behaviour to the dynamics of mass societies. A nation of 100 million people functions in a fundamentally different way to a band of a hundred individuals.[Read more at location 2405](#)

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Threats and promises often succeed in creating stable human hierarchies and mass-cooperation networks, as long as people believe that they reflect the inevitable

laws of nature or the divine commands of God, rather than just human whims. All large-scale human cooperation is ultimately based on our belief in imagined orders. These are sets of rules that, despite existing only in our imagination, we believe to be as real and inviolable as gravity.[Read more at location 2466](#)

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Most people presume that reality is either objective or subjective, and that there is no third option. Hence once they satisfy themselves that something isn't just their own subjective feeling, they jump to the conclusion it must be objective. If lots of people believe in God; if money makes the world go round; and if nationalism starts wars and builds empires – then these things aren't just a subjective belief of mine. God, money and nations must therefore be objective realities. However, there is a third level of reality: the intersubjective level. Intersubjective entities depend on communication among many humans rather than on the beliefs and feelings of individual humans. Many of the most important agents in history are intersubjective. Money, for example, has no objective value. You cannot eat, drink or wear a dollar bill. Yet as long as billions of people believe in its value, you can use it to buy food, beverages and clothing.[Read more at location 2482](#)

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It is relatively easy to accept that money is an intersubjective reality. Most people are also happy to acknowledge that ancient Greek gods, evil empires and the values of alien cultures exist only in the imagination. Yet we don't want to accept that our God, our nation or our values are mere fictions, because these are the things that give meaning to our lives. We want to believe that our lives have some objective meaning, and that our sacrifices matter to something beyond the stories in our head. Yet in truth the lives of most people have meaning only within the network of stories they tell one another.[Read more at location 2507](#)

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Meaning is created when many people weave together a common network of stories. Why does a particular action – such as getting married in church, fasting on Ramadan or voting on election day – seem meaningful to me? Because my parents also think it is meaningful, as do my brothers, my neighbours, people in nearby cities and even the residents of far-off countries. And why do all these people think it is meaningful? Because their friends and neighbours also share the same view. People constantly reinforce each other's beliefs in a self-perpetuating loop. Each round of mutual confirmation tightens the web of meaning further, until you have little choice but to believe what everyone else believes. Yet over decades and centuries the web of meaning unravels and a new web is spun in its place. To study history means to watch the spinning and unravelling of these webs, and to realise that what seems to people in one age the most important thing in life becomes utterly meaningless to their descendants.[Read more at location 2515](#)

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Sapiens rule the world because only they can weave an intersubjective web of meaning: a web of laws, forces, entities and places that exist purely in their common imagination. This web allows humans alone to organise crusades, socialist revolutions and human rights movements.[Read more at location 2579](#)

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Maybe someday breakthroughs in neurobiology will enable us to explain communism and the crusades in strictly biochemical terms. Yet we are very far from that point. During the twenty-first century the border between history and biology is likely to blur not because we will discover biological explanations for historical events, but rather because ideological fictions will rewrite DNA strands; political and economic interests will redesign the climate; and the geography of mountains and rivers will give way to cyberspace. As human fictions are translated into genetic and electronic codes, the intersubjective reality will swallow up the objective reality and biology will merge with history. In the twenty-first century fiction might thereby become the most potent force on earth, surpassing even wayward asteroids and natural selection. Hence if we want to understand our future, cracking genomes and crunching numbers is hardly enough. We must also decipher the fictions that give meaning to the world.[Read more at location 2604](#)

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In illiterate societies people make all calculations and decisions in their heads. In literate societies people are organised into networks, so that each person is only a small step in a huge algorithm, and it is the algorithm as a whole that makes the important decisions. This is the essence of bureaucracy.[Read more at location 2704](#)

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Yet even though Herodotus and Thucydides understood reality much better than the authors of the Bible, when the two world views collided, the Bible won by a knockout. The Greeks adopted the Jewish view of history, rather than vice versa. A thousand years after Thucydides, the Greeks became convinced that if some barbarian horde invaded, surely it was divine punishment for their sins. No matter how mistaken the biblical world view was, it provided a better basis for large-scale human cooperation.[Read more at location 2917](#)

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When examining the history of any human network, it is therefore advisable to stop from time to time and look at things from the perspective of some real entity. How do you know if an entity is real? Very simple – just ask yourself, ‘Can it suffer?’ When people burn down the temple of Zeus, Zeus doesn’t suffer. When the euro loses its value, the euro doesn’t suffer. When a bank goes bankrupt, the bank doesn’t suffer. When a country suffers a defeat in war, the country doesn’t really suffer. It’s just a metaphor. In contrast, when a soldier is wounded in battle, he really does suffer. When a famished peasant has nothing to eat, she suffers. When a cow is separated from her newborn calf, she suffers. This is reality.[Read more at location 2962](#)

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Modern science certainly changed the rules of the game, but it did not simply replace myths with facts. Myths continue to dominate humankind. Science only makes these myths stronger. Instead of destroying the intersubjective reality, science will enable it to control the objective and subjective realities more completely than ever before. Thanks to computers and bioengineering, the difference between fiction and reality will blur, as people reshape reality to match their pet fictions.[Read more at location 2998](#)

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From a historical perspective, the spiritual journey is always tragic, for it is a lonely path fit for individuals rather than for entire societies. Human cooperation requires firm answers rather than just questions, and those who foam against stultified religious structures end up forging new structures in their place. It happened to the dualists, whose spiritual journeys became religious establishments. It happened to Martin Luther, who after challenging the laws, institutions and rituals of the Catholic Church found himself writing new law books, founding new institutions and inventing new ceremonies. It happened even to Buddha and Jesus. In their uncompromising quest for the truth they subverted the laws, rituals and structures of traditional Hinduism and Judaism. But eventually more laws, more rituals and more structures were created in their name than in the name of any other person in history.[Read more at location 3119](#)

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Up until modern times, most cultures believed that humans play a part in some great cosmic plan. The plan was devised by the omnipotent gods, or by the eternal laws of nature, and humankind could not change it. The cosmic plan gave meaning to human life, but also restricted human power.[Read more at location 3327](#)

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The modern world does not believe in purpose, only in cause. If modernity has a motto, it is ‘shit happens’. On the other hand, if shit just happens, without any binding script or purpose, then humans too are not limited to any predetermined role. We can do anything we want – provided we can find a way. We are constrained by nothing except our own ignorance. Plagues and droughts have no cosmic meaning – but we can eradicate them. Wars are not a necessary evil on the way to a better future – but we can make peace. No paradise awaits us after death – but we can create paradise here on earth, and live in it for ever, if we just manage to overcome some technical difficulties.[Read more at location 3345](#)

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The modern deal thus offers humans an enormous temptation, coupled with a colossal threat. Omnipotence is in front of us, almost within our reach, but below us yawns the abyss of complete nothingness. On the practical level, modern life consists of a constant pursuit of power within a universe devoid of meaning. Modern culture is the most powerful in history, and it is ceaselessly researching, inventing, discovering and growing. At the same time, it is plagued by more existential angst than any previous culture.[Read more at location 3355](#)

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Traditional religions such as Christianity and Islam sought ways to solve humanity’s problems with the help of current resources, either by redistributing the existing pie, or by promising us a pie in the sky. Modernity, in contrast, is based on the firm belief that

economic growth is not only possible but is absolutely essential. Prayers, good deeds and meditation can be comforting and inspiring, but problems such as famine, plague and war can only be solved through growth. This fundamental dogma can be summarised in one simple idea: 'If you have a problem, you probably need more stuff, and in order to have more stuff, you must produce more of it.'[Read more at location 3417](#)

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Nowadays it is generally accepted that some version of free-market capitalism is a much more efficient way of ensuring long-term growth, hence rich farmers and freedom of expression are protected, but ecological habitats, social structures and traditional values that stand in the way of free-market capitalism are destroyed and dismantled.[Read more at location 3465](#)

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Capitalism did make an important contribution to global harmony by encouraging people to stop viewing the economy as a zero-sum game, in which your profit is my loss, and instead see it as a win-win situation, in which your profit is also my profit. This has probably helped global harmony far more than centuries of Christian preaching about loving your neighbour and turning the other cheek.[Read more at location 3483](#)

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The traditional view of the world as a pie of a fixed size presupposes there are only two kinds of resources in the world: raw materials and energy. But in truth, there are three kinds of resources: raw materials, energy and knowledge. Raw materials and energy are exhaustible – the more you use, the less you have. Knowledge, in contrast, is a growing resource – the more you use, the more you have.[Read more at location 3523](#)

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Humankind finds itself locked into a double race. On the one hand, we feel compelled to speed up the pace of scientific progress and economic growth. A billion Chinese and a billion Indians want to live like middle-class Americans, and they see no reason why they should put their dreams on hold when the Americans are unwilling to give up their SUVs and shopping malls. On the other hand, we must stay at least one step ahead of ecological Armageddon. Managing this double race becomes more difficult by the year, because every stride that brings the Delhi slum-dwellers closer to the American Dream also brings the planet closer to the brink.[Read more at location 3560](#)

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#How rational is it to risk the future of humankind on the assumption that future scientists will make some unknown discoveries? Most of the presidents, ministers and CEOs who run the world are very rational people. Why are they willing to take such a gamble? Maybe because they don't think they are gambling on their own personal future. Even if bad comes to worse and science cannot hold off the deluge, engineers could still build a hi-tech Noah's Ark for the upper caste, while leaving billions of others to drown.[Read more at location 3594](#)

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The great political, artistic and religious project of modernity has been to find a meaning to life that is not rooted in some great cosmic plan. We are not actors in a divine drama, and nobody cares about us and our deeds, so nobody sets limits to our power – but we are still convinced our lives have meaning.[Read more at location 3660](#)

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Throughout history prophets and philosophers have argued that if humans stopped believing in a great cosmic plan, all law and order would vanish. Yet today, those who pose the greatest threat to global law and order are precisely those people who continue to believe in God and His all-encompassing plans. God-fearing Syria is a far more violent place than the atheist Netherlands.[Read more at location 3663](#)

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The humanist religion worships humanity, and expects humanity to play the part that God played in Christianity and Islam, and that the laws of nature played in Buddhism and Daoism. Whereas traditionally the great cosmic plan gave meaning to the life of humans, humanism reverses the roles, and expects the experiences of humans to give meaning to the great cosmos. According to humanism, humans must draw from within their inner experiences not only the meaning of their own lives, but also the meaning of the entire universe. This is the primary commandment humanism has given us: create meaning for a meaningless world.[Read more at location 3671](#)

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We aren't born with a ready-made conscience. As we pass through life, we hurt people and people hurt us, we act compassionately and others show compassion to us. If we pay attention, our moral sensitivity sharpens, and these experiences become a source of valuable ethical knowledge about what is good, what is right and who I really am. Humanism thus sees life as a gradual process of inner change, leading from ignorance to enlightenment by means of experiences. The highest aim of humanist life is to fully develop your knowledge through a large variety of intellectual, emotional and physical experiences.[Read more at location 3928](#)

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Democratic elections usually work only within populations that have some prior common bond, such as shared religious beliefs and national myths. They are a method to settle disagreements between people who already agree on the basics. Accordingly, in many cases liberalism has fused with age-old collective identities and tribal feelings to form modern nationalism.[Read more at location 4095](#)

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Despite liberalising its politics and economics, China is neither a democracy nor a truly free-market economy, which does not prevent it from becoming the economic giant of the twenty-first century. Yet this economic giant casts a very small ideological shadow. Nobody seems to know what the Chinese believe these days – including the Chinese themselves. In theory China is still communist, but in practice it is nothing of the kind. Some Chinese thinkers and leaders toy with a return to Confucianism, but that's hardly more than a convenient veneer. This ideological vacuum makes China the most promising breeding ground for the new techno-religions emerging from Silicon Valley (which we will discuss in the following chapters). But these techno-religions, with their belief in immortality and virtual paradises, would take at least a decade or two to establish themselves. Hence at present, China doesn't pose a real alternative to liberalism. If bankrupted Greeks despair of the liberal model and search for a substitute, 'imitating the Chinese' doesn't mean much.[Read more at location 4387](#)

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Liberals value individual liberty so much because they believe that humans have free will. According to liberalism, the decisions of voters and customers are neither deterministic nor random. People are of course influenced by external forces and chance events, but at the end of the day each of us can wave the magic wand of freedom and decide things for ourselves. This is the reason liberalism gives so much importance to voters and customers, and instructs us to follow our heart and do what feels good. It is our free will that imbues the universe with meaning, and since no outsider can know how you really feel or predict your choices for sure, you shouldn't trust any Big Brother to look after your interests and desires.[Read more at location 4562](#)

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The contradiction between free will and contemporary science is the elephant in the laboratory, whom many prefer not to see as they peer into their microscopes and fMRI scanners.¹[Read more at location 4570](#)

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Doubting free will is not just a philosophical exercise. It has practical implications. If organisms indeed lack free will, it implies we could manipulate and even control their desires using drugs, genetic engineering or direct brain stimulation.[Read more at location 4636](#)

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Sally Adee, a journalist for the New Scientist, was allowed to visit a training facility for snipers and test the effects herself. At first, she entered a battlefield simulator without

wearing the transcranial helmet. Sally describes how fear swept over her as she saw twenty masked men, strapped with suicide bombs and armed with rifles, charge straight towards her. 'For every one I manage to shoot dead,' writes Sally, 'three new assailants pop up from nowhere. I'm clearly not shooting fast enough, and panic and incompetence are making me continually jam my rifle.' Luckily for her, the assailants were just video images, projected on huge screens all around her. Still, she was so disappointed with her poor performance that she felt like putting down the rifle and leaving the simulator. Then they wired her up to the helmet. She reports feeling nothing unusual, except a slight tingle and a strange metallic taste in her mouth. Yet she began picking off the terrorists one by one, as coolly and methodically as if she were Rambo or Clint Eastwood. 'As twenty of them run at me brandishing their guns, I calmly line up my rifle, take a moment to breathe deeply, and pick off the closest one, before tranquilly assessing my next target. In what seems like next to no time, I hear a voice call out, "Okay, that's it." The lights come up in the simulation room . . . In the sudden quiet amid the bodies around me, I was really expecting more assailants, and I'm a bit disappointed when the team begins to remove my electrodes. I look up and wonder if someone wound the clocks forward. Inexplicably, twenty minutes have just passed. "How many did I get?" I ask the assistant. She looks at me quizzically. "All of them." The experiment changed Sally's life. In the following days she realised she has been through a 'near-spiritual experience . . . what defined the experience was not feeling smarter or learning faster: the thing that made the earth drop out from under my feet was that for the first time in my life, everything in my head finally shut up . . . My brain without self-doubt was a revelation. There was suddenly this incredible silence in my head . . . I hope you can sympathise with me when I tell you that the thing I wanted most acutely for the weeks following my experience was to go back and strap on those electrodes. I also started to have a lot of questions. Who was I apart from the angry bitter gnomes that populate my mind and drive me to failure because I'm too scared to try? And where did those voices come from?'⁷ Some of those voices repeat society's prejudices, some echo our personal history, and some articulate our genetic legacy. All of them together, says Sally, create an invisible story that shapes our conscious decisions in ways we seldom grasp. What would happen if we could rewrite our inner monologues, or even silence them completely on occasion?[Read more at location 4675](#)

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Every time the narrating self evaluates our experiences, it discounts their duration, and adopts the 'peak-end rule' – it remembers only the peak moment and the end moment, and evaluates the whole experience according to their average. This has far-reaching impact on all our practical decisions.[Read more at location 4797](#)

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Most people identify with their narrating self. When they say 'I', they mean the story in their head, not the stream of experiences they undergo. We identify with the inner system that takes the crazy chaos of life and spins out of it seemingly logical and consistent yarns. It doesn't matter that the plot is full of lies and lacunas, and that it is rewritten again and again, so that today's story flatly contradicts yesterday's; the important thing is that we always retain the feeling that we have a single unchanging identity from birth to death (and perhaps even beyond the grave). This gives rise to the questionable liberal belief that I am an individual, and that I possess a consistent and clear inner voice, which provides meaning for the entire universe.¹⁸[Read more at location 4853](#)

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You want to make people believe in imaginary entities such as gods and nations, you should make them sacrifice something valuable. The more painful the sacrifice, the more convinced people are of the existence of the imaginary recipient.[Read more at location 4901](#)

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The self too is an imaginary story, just like nations, gods and money. Each of us has a sophisticated system that throws away most of our experiences, keeps only a few choice samples, mixes them up with bits from movies we saw, novels we read, speeches we heard, and from our own daydreams, and weaves out of all that jumble a seemingly coherent story about who I am, where I came from and where I am going. This story tells me what to love, whom to hate and what to do with myself. This story may even cause me to sacrifice my life, if that's what the plot requires. We all have our genre. Some people live a tragedy, others inhabit a never-ending religious drama, some approach life as if it were an action film, and not a few act as if in a comedy. But in the end, they are all just stories.[Read more at location 4924](#)

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Liberals uphold free markets and democratic elections because they believe that every human is a uniquely valuable individual, whose free choices are the ultimate source of authority. In the twenty-first century three practical developments might make this belief obsolete: 1. Humans will lose their economic and military usefulness, hence the economic and political system will stop attaching much value to them. 2. The system will still find value in humans collectively, but not in unique individuals. 3. The system will still find value in some unique individuals, but these will be a new elite of upgraded superhumans rather than the mass of the population.[Read more at location 4958](#)

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Computers function very differently from humans, and it seems unlikely that computers will become humanlike any time soon. In particular, it doesn't seem that computers are about to gain consciousness, and to start experiencing emotions and sensations. Over the last decades there has been an immense advance in computer intelligence, but there has been exactly zero advance in computer consciousness. As far as we know, computers in 2016 are no more conscious than their prototypes in the 1950s. However, we are on the brink of a momentous revolution. Humans are in danger of losing their value, because intelligence is decoupling from consciousness.[Read more at location 5022](#)

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The most important question in twenty-first-century economics may well be what to do with all the superfluous people. What will conscious humans do, once we have highly intelligent non-conscious algorithms that can do almost everything better?[Read more at location 5136](#)

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In the nineteenth century the Industrial Revolution created a huge new class of urban proletariats, and socialism spread because no one else managed to answer their unprecedented needs, hopes and fears. Liberalism eventually defeated socialism only by adopting the best parts of the socialist programme. In the twenty-first century we might witness the creation of a new massive class: people devoid of any economic, political or even artistic value, who contribute nothing to the prosperity, power and glory of society.[Read more at location 5249](#)

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The crucial problem isn't creating new jobs. The crucial problem is creating new jobs that humans perform better than algorithms.²⁰ The technological bonanza will probably make it feasible to feed and support the useless masses even without any effort on their side. But what will keep them occupied and content? People must do something, or they will go crazy. What will they do all day? One solution might be offered by drugs and computer games. Unnecessary people might spend increasing amounts of time within 3D virtual-reality worlds, which would provide them with far more excitement and emotional engagement than the drab reality outside. Yet such a development would deal a mortal blow to the liberal belief in the sacredness of human life and of human experiences. What's so sacred in useless bums who pass their days devouring artificial experiences in La La Land?[Read more at location 5268](#)

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It would be extremely difficult for humans to control the motivation of a system smarter than themselves. Even preprogramming the system with seemingly benign goals might backfire horribly. One popular scenario imagines a corporation designing the first artificial super-intelligence, and giving it an innocent test such as calculating pi. Before anyone realises what is happening, the AI takes over the planet, eliminates the human race, launches a conquest campaign to the ends of the galaxy, and transforms the entire known universe into a giant super-computer that for billions upon billions of years calculates pi ever more accurately. After all, this is the divine mission its Creator gave it.[Read more at location 5278](#)

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The algorithms won't revolt and enslave us. Rather, the algorithms will be so good in making decisions for us that it would be madness not to follow their advice.[Read more at location 5394](#)

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Liberalism sanctifies the narrating self, and allows it to vote in the polling stations, in the supermarket and in the marriage market. For centuries this made good sense, because though the narrating self believed in all kinds of fictions and fantasies, no alternative system knew me better. Yet once we have a system that really does know me better, it will be foolhardy to leave authority in the hands of the narrating self.[Read more at location 5464](#)

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A recent study commissioned by Google's nemesis – Facebook – has indicated that already today the Facebook algorithm is a better judge of human personalities and dispositions even than people's friends, parents and spouses. The study was conducted on 86,220 volunteers who have a Facebook account and who completed a hundred-item personality questionnaire. The Facebook algorithm predicted the volunteers' answers based on monitoring their Facebook Likes – which webpages, images and clips they tagged with the Like button. The more Likes, the more accurate the predictions. The algorithm's predictions were compared with those of work colleagues, friends, family members and spouses. Amazingly, the algorithm needed a set of only ten Likes in order to outperform the predictions of work colleagues. It needed seventy Likes to outperform friends, 150 Likes to outperform family members and 300 Likes to outperform spouses. In other words, if you happen to have clicked 300 Likes on your Facebook account, the Facebook algorithm can predict your opinions and desires better than your husband or wife![Read more at location 5490](#)

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The shifting of authority from humans to algorithms is happening all around us, not as a result of some momentous governmental decision, but due to a flood of mundane choices. The result will not be an Orwellian police state. We always prepare ourselves for the previous enemy, even when we face an altogether new menace. Defenders of human individuality stand guard against the tyranny of the collective, without realising that human individuality is now threatened from the opposite direction. The individual will not be crushed by Big Brother; it will disintegrate from within. Today corporations and governments pay homage to my individuality, and promise to provide medicine, education and entertainment customised to my unique needs and wishes. But in order to do so, corporations and governments first need to break me up into biochemical subsystems, monitor these subsystems with ubiquitous sensors and decipher their working with powerful algorithms. In the process, the individual will transpire to be nothing but a religious fantasy. Reality will be a mesh of biochemical and electronic algorithms, without clear borders, and without individual hubs.[Read more at location 5579](#)

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The third threat to liberalism is that some people will remain both indispensable and undecipherable, but they will constitute a small and privileged elite of upgraded humans. These superhumans will enjoy unheard-of abilities and unprecedented creativity, which will allow them to go on making many of the most important decisions in the world. They will perform crucial services for the system, while the system could not understand and manage them. However, most humans will not be upgraded, and they will consequently become an inferior caste, dominated by both computer algorithms and the new superhumans.[Read more at location 5594](#)

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Twentieth-century medicine aimed to heal the sick. Twenty-first-century medicine is increasingly aiming to upgrade the healthy. Healing the sick was an egalitarian project, because it assumed that there is a normative standard of physical and mental health that everyone can and should enjoy. If someone fell below the norm, it was the job of doctors to fix the problem and help him or her 'be like everyone'. In contrast, upgrading the healthy is an elitist project, because it rejects the idea of a universal standard applicable to all, and seeks to give some individuals an edge over others. People want superior memories, above-average intelligence and first-class sexual abilities. If some form of upgrade becomes so cheap and common that everyone enjoys it, it will simply be considered the new baseline, which the next generation of treatments will strive to surpass.[Read more at location 5626](#)

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Unlike in the twentieth century, when the elite had a stake in fixing the problems of the poor because they were militarily and economically vital, in the twenty-first century the most efficient (albeit ruthless) strategy may be to let go of the useless third-class carriages, and dash forward with the first class only.[Read more at location 5647](#)

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The great human projects of the twentieth century – overcoming famine, plague and war – aimed to safeguard a universal norm of abundance, health and peace for all people without exception. The new projects of the twenty-first century – gaining immortality, bliss and divinity – also hope to serve the whole of humankind. However, because these projects aim at surpassing rather than safeguarding the norm, they may well result in the creation of a new superhuman caste that will abandon its liberal roots and treat normal

humans no better than nineteenth-century Europeans treated Africans.[Read more at location 5654](#)

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We may successfully upgrade our bodies and our brains, while losing our minds in the process. Indeed, techno-humanism may end up downgrading humans. The system may prefer downgraded humans not because they would possess any superhuman knacks, but because they would lack some really disturbing human qualities that hamper the system and slow it down.[Read more at location 5847](#)

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Techno-humanism faces an impossible dilemma here. It considers the human will to be the most important thing in the universe, hence it pushes humankind to develop technologies that can control and redesign our will. After all, it's tempting to gain control over the most important thing in the world. Yet once we have such control, techno-humanism would not know what to do with it, because the sacred human will would become just another designer product. We can never deal with such technologies as long as we believe that the human will and the human experience are the supreme source of authority and meaning.[Read more at location 5899](#)

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Dataism says that the universe consists of data flows, and the value of any phenomenon or entity is determined by its contribution to data processing.¹ This may strike you as some eccentric fringe notion, but in fact it has already conquered most of the scientific establishment. Dataism was born from the explosive confluence of two scientific tidal waves. In the 150 years since Charles Darwin published *On the Origin of Species*, the life sciences have come to see organisms as biochemical algorithms. Simultaneously, in the eight decades since Alan Turing formulated the idea of a Turing Machine, computer scientists have learned to engineer increasingly sophisticated electronic algorithms. Dataism puts the two together, pointing out that exactly the same mathematical laws apply to both biochemical and electronic algorithms. Dataism thereby collapses the barrier between animals and machines, and expects electronic algorithms to eventually decipher and outperform biochemical algorithms. For politicians, business people and ordinary consumers, Dataism offers groundbreaking technologies and immense new powers. For scholars and intellectuals it also promises to provide the scientific holy grail that has eluded us for centuries: a single overarching theory that unifies all the scientific disciplines from literature and musicology to economics and biology.[Read more at location 5910](#)

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Capitalism did not defeat communism because capitalism was more ethical, because individual liberties are sacred or because God was angry with the heathen communists. Rather, capitalism won the Cold War because distributed data processing works better than centralised data processing, at least in periods of accelerating technological changes.[Read more at location 5985](#)

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In the coming decades, it is likely that we will see more Internet-like revolutions, in which technology steals a march on politics. Artificial intelligence and biotechnology might soon overhaul our societies and economies – and our bodies and minds too – but they are hardly a blip on our political radar. Our current democratic structures just cannot collect and process the relevant data fast enough, and most voters don't understand biology and cybernetics well enough to form any pertinent opinions. Hence traditional democratic politics loses control of events, and fails to provide us with meaningful visions for the future.[Read more at location 6028](#)

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Precisely because technology is now moving so fast, and parliaments and dictators alike are overwhelmed by data they cannot process quickly enough, present-day politicians are thinking on a far smaller scale than their predecessors a century ago. In the early twenty-first century, politics is consequently bereft of grand visions. Government has become mere administration. It manages the country, but it no longer leads.[Read more at location 6046](#)

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To some extent, this is a very good thing. Given that some of the big political visions of the twentieth century led us to Auschwitz, Hiroshima and the Great Leap Forward, maybe we are better off in the hands of petty-minded bureaucrats. Mixing godlike technology with megalomaniac politics is a recipe for disaster. Many neo-liberal economists and political scientists argue that it is best to leave all the important

decisions in the hands of the free market. They thereby give politicians the perfect excuse for inaction and ignorance, which are reinterpreted as profound wisdom. Politicians find it convenient to believe that the reason they don't understand the world is that they need not understand it. Yet mixing godlike technology with myopic politics also has its downside. Lack of vision isn't always a blessing, and not all visions are necessarily bad. In the twentieth century, the dystopian Nazi vision did not fall apart spontaneously. It was defeated by the equally grand visions of socialism and liberalism. It is dangerous to trust our future to market forces, because these forces do what's good for the market rather than what's good for humankind or for the world. The hand of the market is blind as well as invisible, and left to its own devices it may fail to do anything about the threat of global warming or the dangerous potential of artificial intelligence.[Read more at location 6050](#)

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As the global data-processing system becomes all-knowing and all-powerful, so connecting to the system becomes the source of all meaning. Humans want to merge into the data flow because when you are part of the data flow you are part of something much bigger than yourself. Traditional religions told you that your every word and action was part of some great cosmic plan, and that God watched you every minute and cared about all your thoughts and feelings. Data religion now says that your every word and action is part of the great data flow, that the algorithms are constantly watching you and that they care about everything you do and feel. Most people like this very much. For true-believers, to be disconnected from the data flow risks losing the very meaning of life. What's the point of doing or experiencing anything if nobody knows about it, and if it doesn't contribute something to the global exchange of information?[Read more at location 6208](#)

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Dataism adopts a strictly functional approach to humanity, appraising the value of human experiences according to their function in data-processing mechanisms. If we develop an algorithm that fulfils the same function better, human experiences will lose their value. Thus if we can replace not just taxi drivers and doctors but also lawyers, poets and musicians with superior computer programs, why should we care if these programs have no consciousness and no subjective experiences? If some humanist starts adulating the sacredness of human experience, Dataists would dismiss such sentimental humbug.[Read more at location 6249](#)

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By equating the human experience with data patterns, Dataism undermines our main source of authority and meaning, and heralds a tremendous religious revolution, the like of which has not been seen since the eighteenth century. In the days of Locke, Hume and Voltaire humanists argued that 'God is a product of the human imagination'. Dataism now gives humanists a taste of their own medicine, and tells them: 'Yes, God is a product of the human imagination, but human imagination in turn is the product of biochemical algorithms.' In the eighteenth century, humanism sidelined God by shifting from a deo-centric to a homo-centric world view. In the twenty-first century, Dataism may sideline humans by shifting from a homo-centric to a data-centric view.[Read more at location 6264](#)

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Dataism naturally has its critics and heretics. As we saw in Chapter 3, it's doubtful whether life can really be reduced to data flows. In particular, at present we have no idea how or why data flows could produce consciousness and subjective experiences. Maybe we'll have a good explanation in twenty years. But maybe we'll discover that organisms aren't algorithms after all.[Read more at location 6329](#)

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A critical examination of the Dataist dogma is likely to be not only the greatest scientific challenge of the twenty-first century, but also the most urgent political and economic project. Scholars in the life sciences and social sciences should ask themselves whether we miss anything when we understand life as data processing and decision-making. Is there perhaps something in the universe that cannot be reduced to data? Suppose non-conscious algorithms could eventually outperform conscious intelligence in all known data-processing tasks – what, if anything, would be lost by replacing conscious intelligence with superior non-conscious algorithms?[Read more at location 6336](#)

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If Dataism succeeds in conquering the world, what will happen to us humans? In the beginning, it will probably accelerate the humanist pursuit of health, happiness and

power. Dataism spreads itself by promising to fulfil these humanist aspirations. In order to gain immortality, bliss and divine powers of creation, we need to process immense amounts of data, far beyond the capacity of the human brain. So the algorithms will do it for us. Yet once authority shifts from humans to algorithms, the humanist projects may become irrelevant. Once we abandon the homo-centric world view in favour of a data-centric world view, human health and happiness may seem far less important.[Read more at location 6347](#)

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The world is changing faster than ever before, and we are flooded by impossible amounts of data, of ideas, of promises and of threats. Humans relinquish authority to the free market, to crowd wisdom and to external algorithms partly because they cannot deal with the deluge of data. In the past, censorship worked by blocking the flow of information. In the twenty-first century, censorship works by flooding people with irrelevant information. People just don't know what to pay attention to, and they often spend their time investigating and debating side issues. In ancient times having power meant having access to data. Today having power means knowing what to ignore.[Read more at location 6374](#)

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If we think in term of months, we had probably focus on immediate problems such as the turmoil in the Middle East, the refugee crisis in Europe and the slowing of the Chinese economy. If we think in terms of decades, then global warming, growing inequality and the disruption of the job market loom large. Yet if we take the really grand view of life, all other problems and developments are overshadowed by three interlinked processes: 1. Science is converging on an all-encompassing dogma, which says that organisms are algorithms, and life is data processing. 2. Intelligence is decoupling from consciousness. 3. Non-conscious but highly intelligent algorithms may soon know us better than we know ourselves. These three processes raise three key questions, which I hope will stick in your mind long after you have finished this book: 1.

Are organisms really just algorithms, and is life really just data processing? 2. What's more valuable – intelligence or consciousness? 3. What will happen to society, politics and daily life when non-conscious but highly intelligent algorithms know us better than we know ourselves?